

Spatial and temporal dynamics of the endothelium.

The endothelium is a highly metabolically active organ that is involved in many physiological processes, including the control of vasomotor tone, barrier function, leukocyte adhesion and trafficking, inflammation, and hemostasis. Endothelial cell phenotypes are differentially regulated in space and time. Endothelial cell heterogeneity has important implications for developing strategies in basic research, diagnostics and therapeutics. The goals of this review are to: (i) consider mechanisms of endothelial cell heterogeneity; (ii) discuss the bench-to-bedside gap in endothelial biomedicine; (iii) revisit definitions for endothelial cell activation and dysfunction; and (iv) propose new goals in diagnosis and therapy. Finally, these themes will be applied to an understanding of vascular bed-specific hemostasis.